

Core Gameplay Loop & Retention Design

How steps should power the game — research findings and recommendation · July 12, 2026 · Status: **direction approved** · tuning v1

TL;DR

Neither pure model wins. One-attack-per-day as the *only* mechanic wastes most of your steps' emotional value. A pure 24/7 idle brawl with a hard "your character can't fight" cutoff is the single most reliable churn mechanic in health apps.

Recommendation — a hybrid: steps are *fuel* for a character who fights around the clock; the daily DEPLOY stays as the big appointment moment; and high step counts charge a temporary "Overdrive" special-attack mode. The one change from the original sketch: when someone runs out of fuel, their character *rests at camp* — it is never punished, and the party never suffers for it.

1 · The question

The app currently uses a **bank-and-burst** model: steps become banked Energy, spent in one manual DEPLOY per day, plus a modest idle damage drip (capped at 10 hours offline). The question: should it instead move toward a **fuel-the-fighter** model — the character fights the boss 24/7, steps are the fuel, running dry stops the character, and big step totals unlock a special-attack mode?

Judged against two goals that must both hold: **players keep coming back**, and **the app makes a friend group healthier** — meaning a low-motivation week must always be survivable.

2 · What the research says

Findings from step-tracking games (Pokémon GO, Pikmin Bloom, Walkr, Wokamon, Habitica), idle games (AFK Arena, Idle Heroes, Egg Inc., Cookie Clicker), habit apps (Duolingo), and academic work on gamified fitness:

Fitness apps lose ~80% of users within two weeks. The steepest churn curve of any consumer app category. Every mechanic should be judged by: *does this make a bad week survivable?*

Punishment churns; pause retains. Habitica — the closest analogue to Stride Quest (party members' habits damage a shared boss) — is the cautionary tale: missed days deal damage to you *and your party*, and it is the app's most-documented source of anxiety, avoidance, and quitting "when life gets hard." Every successful step game (Pokémon GO, Pikmin Bloom, Walkr, Wokamon) simply *pauses* when you don't walk. Nothing decays; nothing dies.

Loss aversion works — but only on bonuses, never on owned progress. A 2016 clinical trial (Annals of Internal Medicine) found people hit step goals significantly more when framed as losing a held bonus; the equivalent gain-framed reward did not work. Duolingo's streak is the strongest retention lever in mobile ("7 days I could lose") — yet its biggest retention *wins* came from softening the loss: auto-applied streak freezes cut at-risk churn by 21%. Unforgiving streaks turn one sick day into "why bother restarting."

Idle games retain through the collect-and-spend ritual. Every session opens with a celebration (collect what accrued), flows into an obvious spend, and ends when *the player* runs out of resources — an exit that feels self-chosen, not imposed. Idle games post exceptional retention (60%+ day-1 vs ~40% industry average) on exactly this loop. Offline caps in the 8–12 hour band produce a natural two-check-ins-per-day rhythm — the current 10h cap is already in the sweet spot.

Burst multipliers belong on the active moment, never on idle income. Cookie Clicker's Golden Cookies, Egg Inc.'s timed boosts, Tsum Tsum's fever mode — idle is the floor, attention is the spike. The existing "crits live on deploy, not idle" decision matches this exactly.

For a friend group, the social loop IS the retention engine — but contribution, not obligation. Duolingo friend streaks lift daily completion 22%, and peer nudges outperform app notifications. The dark side: Snapchat-style streak obligation breeds resentment, and Habitica-style "your lapse hurts the party" is toxic. In a 3–8 person group, one person churning can collapse the whole guild — protecting the *least* active member's dignity is paramount.

3 · The three ideas, judged

"Spend steps on one attack a day"

KEEP — AS THE RITUAL, NOT THE GAME

As the sole mechanic it makes the app a 30-second daily errand and renders steps 2 through 15,000 emotionally identical. But as the *anchor moment* layered on an idle floor, it is exactly the "daily dungeon" appointment mechanic that top games use. **Verdict: keep DEPLOY; don't make it everything.**

"Characters fight 24/7, fueled by steps"

YES — THE RIGHT BACKBONE

This is the Walkr/Wokamon model, and it fixes the biggest weakness of bank-and-burst: today, walking at 3pm does nothing visible until tomorrow's deploy. In a fuel model, every walk literally keeps your hero swinging — the battlefield screen becomes a living diorama where the whole party fights at rates reflecting everyone's real activity. It also gives idle accrual a *fiction* (fuel burning) instead of an abstract multiplier — which is where energy systems succeed rather than feel bolted-on.

"Under 2k steps in 2 days → can't fight"

RIGHT INSTINCT, WRONG FRAMING

A hard lockout is the Habitica / Midnight Star mistake ("players don't appreciate being told they must stop"). Reframe it:

- **Fuel low** → hero fights *slower* (soft floor, e.g. 25% rate).
- **Fuel empty** → hero **sits at the campfire and rests**. The animation is cozy, not shameful; no HP loss, no party damage, no red warnings. The comeback story is "your hero wakes up hungry for battle," not "you failed."
- **Best co-op hook in the design:** a teammate can send a bit of their own energy to wake a resting friend — the nudge comes from a friend, not the app, which is precisely the pattern that outperforms push notifications.

"20k in 2 days → special attack mode"

YES — AND THE ART ALREADY EXISTS

A charged **Overdrive** meter that fills on high step volume and unleashes a timed fever mode (the hero chains special-attack animations at a big multiplier for a few hours) is the canonical Golden-Cookie spike: big walking days feel spectacular without making normal days feel broken. Special-attack animations for all 40 jobs already exist in assets/effects/ — this mechanic is the payoff for that investment.

4 • Recommended loop

1. **Steps fill your fuel tank.** Your hero fights the boss continuously, burning fuel — damage rate scales with your Job multiplier, as today.
2. **Open the app anytime → COLLECT** the damage that accrued (celebration first, always) and see the party fighting live.
3. **Once a day → DEPLOY.** The big button: spend banked energy for a burst hit with crits and the streak multiplier — the juice moment and the daily habit anchor.
4. **Sustained high stepping charges OVERDRIVE:** a timed special-attack mode that turns a 20k weekend into a fireworks show.
5. **Run dry → your hero rests at camp.** Zero punishment. A friend can share fuel to wake you.
6. **Streaks keep their multiplier but gain REST TOKENS** — earned by hitting weekly goals, auto-applied like Duolingo's streak freezes — so one sick day never erases two weeks of momentum.
7. **The weekly boss stays** the forgiving co-op arc it already is.

This is deliberately close to what is already built — the fuel/burn framing replaces the abstract idle drip; Overdrive and campfire-rest are new; DEPLOY, streaks, and the weekly boss survive intact. It produces the proven cadence: two 30-second check-ins a day (collect + glance at the party) plus one deliberate session (deploy), with the weekly boss kill as the shared payoff.

Model	Strength	Fatal flaw as a pure model
A — Bank & burst (current)	Strong daily ritual; juicy deploy moment	One 30-second session/day; steps beyond the deploy feel dead; nothing to watch
B — Pure 24/7 idle fuel	Every walk visibly matters; living battle screen	No appointment moment or spike of agency; hard fuel-outs create guilt spirals and churn
C — Hybrid (recommended)	Idle floor + daily anchor + burst spike + weekly arc — the full proven retention stack	More systems to tune (fuel burn rate, overdrive charge)

5 • Two guardrails to write in stone

1 The party is never punished for a member's inactivity.

Boss HP already scales per member; a resting member means slower progress, never backsliding. (Habitica's one proven mistake.)

2 Loss-framing only ever applies to bonuses.

Streak multipliers and overdrive charge can be lost; earned progress, XP, and the character itself never can.

6 · Decisions & calculated tuning (v1 — approved July 12, 2026)

- **Fuel burn rate — calculated: 300 fuel/hour while battling** (1 step = 1 fuel). Anchor: a full 24h of fighting should cost slightly /ess than the 8,000-step daily goal, so goal-hitters build a buffer instead of treading water. $24\text{h} \times 300 = 7,200$ steps; an 8k day banks $\approx +2.7\text{h}$ of surplus. (Exact break-even would be $8,000 \div 24 \approx 333$ — deliberately not chosen.)
- **Overdrive: player-activated.** A saved charge the player pops when they choose — agency makes it feel earned.
- **Rally (fuel-sharing): costs the giver a little** — 500 Energy ($\approx 6\%$ of a goal day). Gifts that cost something carry social weight.

Tunable (starting value)	Value	Why
Fuel burn (Battling)	300 / hour	24h of fighting = 7,200 steps — just under the 8k goal, so goal days bank surplus
Winded state	last 6h of fuel · $\times 0.5$ damage · $\times 0.5$ burn	Soft floor before resting; the final 6 nominal hours stretch to 12 real hours
Resting state	0 damage, 0 burn	Cozy campfire — never punished, no party damage, no red UI
Fuel tank cap	48h (14,400 fuel)	A big weekend carries a player through 2 rest days
Starter fuel	24h (7,200)	A new player's first session never shows a resting hero
Overdrive charge	4,000 steps past the daily goal = 100%	Calibrated to "20k in 2 days": $20\text{k} - 16\text{k goal} = 4\text{k excess}$
Overdrive effect	$\times 3$ idle damage for 4h, special-attack animations	Pure reward (normal burn); spike lives on attention, idle stays the floor
Rally	giver pays 500 Energy → friend gets 6h fuel; 1/day; receiver must be Winded/Resting	A wake-up, not a subsidy; the nudge comes from a friend, not the app
Streak Shields	earn 1 per week at 5 goal-days; hold 2; auto- apply	Duolingo pattern — one sick day never erases two weeks of momentum
Boss base HP	60,000 → 150,000 per member	Output rises under this model (daily crit deploys + 24/7 idle); target kill $\approx$ day 5–6
Offline behavior	burn pauses with the existing 10h accrual cap	Absence pauses the tank, never wastes it

7 · Key sources

- **Duolingo streak system:** Deconstructor of Fun teardown (duolingo.deconstructoroffun.com/mechanics/streaks); Trophy.so case study — streak freeze cut at-risk churn 21%; 7-day-streak users $\sim 3.6\times$ more likely to retain.

- **Habitica punishment criticism:** Habitica wiki (Death Mechanics, Quests/Boss); Trophy.so gamification case study; habi.app on reward-only alternatives.
- **Loss-framing RCT:** Patel et al., *Annals of Internal Medicine* 2016 — loss-framed incentives increased step-goal achievement; gain-framed did not (acpjournals.org/doi/10.7326/M15-1635).
- **Fitness-app churn:** retentioncheck.com fitness benchmarks (~80% loss in 2 weeks; D30 8–12%); Endeavour Partners wearables study (~30% abandon in 6 months).
- **Idle-game session design:** Anthony Pecorella, GDC 2015 "Idle Games: The Mechanics and Monetization" (gdcvault.com/play/1022065); Kolibri Games / GameAnalytics — Idle Miner Tycoon D1 retention 63–81%.
- **Offline caps:** AFK Arena 12h chest; Idle Heroes 8h (~3 logins/day); Egg Inc. silo-capacity caps; Tideward on 24h as daily-habit sweet spot.
- **Fever/burst mechanics:** Cookie Clicker Golden Cookies (Frenzy ×7); Egg Inc. timed boosts; Tsum Tsum fever mode.
- **Energy-system failures:** Mobile Free To Play "Understanding and Eliminating Energy Systems"; Midnight Star postmortem (TouchArcade) on hard lockouts; PocketGamer.biz designer roundtable.
- **Social loops:** Duolingo Friend Streaks (+22% daily completion); Pokémon GO Adventure Sync & buddy system; Walkr fleet Epics; Snapchat streak-obligation studies (ScienceDirect); BeReal DAU decline (Business of Apps).
- **Step-game forgiveness patterns:** Pikmin Bloom weekly challenges (participation prizes even on failure, Niantic support docs); CHI 2022 paper on Pikmin Bloom (dl.acm.org/doi/10.1145/3491101.3519785); "Daily Quests or Daily Pests?" CHI PLAY study on daily-reward fatigue (dl.acm.org/doi/10.1145/3549489).

Prepared July 12, 2026 from two research sweeps (fitness-game retention; idle-game loop design) plus the current game tuning in `convex/gameConfig.ts`. Direction approved July 12, 2026; full spec with config shapes: `docs/superpowers/specs/2026-07-12-core-loop-fuel-hybrid-design.md`. Not yet implemented.